

More examples for 12.8

① Find extreme of $x^2 + 4y^2 - 2x + 8y - 1$ on $\mathbb{R}^2$

$$\nabla f = (2x - 2, 8y + 8) \quad \nabla f = \vec{0} \quad \text{if } x = 1, y = -1$$

$$D = f_{xx} f_{yy} - (f_{xy})^2 = 2 \cdot 8 - 0^2 = 16 > 0 \quad f_{xx} = 2 > 0$$

Local MIN $f(1, -1) = -6$

② Same function on $\frac{x^2}{4} + y^2 \leq 4$

Look at ∂S $\frac{x^2}{4} + y^2 = 4$ $x = 4 \cos \theta$ $y = 2 \sin \theta$

$$f(\theta) = 16 \cos^2 \theta + 16 \sin^2 \theta - 8 \cos \theta + 16 \sin \theta - 1$$

$$= 16 - 8 \cos \theta + 16 \sin \theta - 1$$

$$f'(\theta) = 8 \sin \theta + 16 \cos \theta \quad f' = 0 \Rightarrow 8 \sin \theta + 16 \cos \theta = 0$$

$$8 \sin \theta = -16 \cos \theta$$

$$f(\theta_1) = \cancel{32.88854} - 2.88854$$

$$f(\theta_2) = 32.88854$$

$$\tan \theta = -2$$

$$\theta_1 = -1.10715$$

$$\theta_2 = -1.10715 + \pi$$

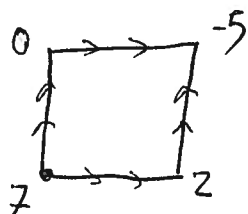
$$= 2.03444$$

③ $x^2 - 6x + y^2 - 8y + 7$ on $\{0 \leq x \leq 1\} \times \{0 \leq y \leq 1\} = S$

$$\nabla f = (2x - 6, 2y - 8) \quad \nabla f = \vec{0} \Rightarrow x = 3 \quad y = 4 \quad \notin S$$

Look at ∂S

$x = 0$	$0 \leq y \leq 1$	$f(0, y) = y^2 - 8y + 7$	$f'(0, y) = 2y - 8 < 0$ on $[0, 1]$
$x = 1$	$0 \leq y \leq 1$	$f(1, y) = y^2 - 8y + 2$	$f'(1, y) = 2y - 8 < 0$ on $[0, 1]$
$y = 0$	$0 \leq x \leq 1$	$f(x, 0) = x^2 - 6x + 7$	$f'(x, 0) = 2x - 6 < 0$ on $[0, 1]$
$y = 1$	$0 \leq x \leq 1$	$f(x, 1) = x^2 - 6x + 1$	$f'(x, 1) = 2x - 6 < 0$ on $[0, 1]$



$$\boxed{f(0,0) = 7}$$

MAX

$$f(0,1) = 0$$

$$f(1,0) = 2$$

$$\boxed{f(1,1) = -5}$$

MIN

§ 12.9 Lagrange Multipliers

Thm To maximize or minimize $f(P)$

subject to the constraint $g(P) = 0$

$$\text{solve } \begin{cases} \nabla f(P) = \lambda \nabla g(P) \\ g(P) = 0 \end{cases}$$

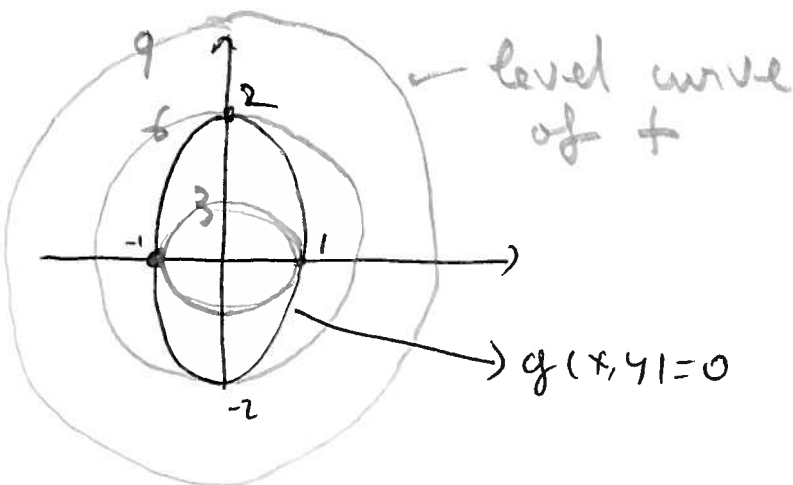
$$\begin{aligned} \text{Ex } f(x, y) &= 2 + x^2 y^2 & \nabla f &= (2x, 2y) \\ g(x, y) &= x^2 + \frac{1}{4}y^2 - 1 = 0 & \nabla g &= (2x, \frac{y}{2}) \end{aligned}$$

$$\begin{cases} 2x = \lambda 2x \\ 2y = \lambda \frac{y}{2} \\ x^2 + \frac{y^2}{4} = 1 \end{cases} \Rightarrow \begin{cases} x=0 \text{ or } x=1 & 2x(1-\lambda)=0 \\ & y(2-\lambda/2)=0 \\ & x^2 + \frac{y^2}{4} = 1 \end{cases}$$

$$x=0 \Rightarrow y^2 = 4 \quad y = \pm 2 \Rightarrow \lambda = 4 \quad (0, \pm 2)$$

$$x \neq 0 \Rightarrow \lambda = 1 \Rightarrow 2 - \frac{\lambda}{2} \neq 0 \Rightarrow y = 0 \Rightarrow x = \pm 1 \quad (\pm 1, 0)$$

$$f(0, 2) = f(0, -2) = 6 \quad f(1, 0) = f(-1, 0) = 3$$



$\nabla f = \lambda \nabla g$ means
that $g(x, y) = 0$ is
tangent to level
curve of f .

other way: $x^2 + \frac{y^2}{4} = 1$ $x = \cos \theta$ $y = 2 \sin \theta$

$$f(\theta) = 2 + \cos^2 \theta + 4 \sin^2 \theta = 3 + 3 \sin^2 \theta$$

$$f'(\theta) = 6 \sin \theta \cos \theta \quad f'(\theta) = 0 \Rightarrow \theta = 0, \pi, \pi/2, 3\pi/2$$

$$f(0) = f(\pi) = 3 \quad f(\pi/2) = f(3\pi/2) = 6$$

Ex $f(x, y) = x^2 + 4y^2 - 2x + 8y - 1$ $\nabla f = (2x-2, 8y+8)$

$$g(x, y) = \frac{x^2}{4} + y^2 - 4 = 0$$

$$\nabla g = (\frac{x}{2}, 2y)$$

$$\begin{cases} 2x-2 = \lambda \frac{x}{2} \\ 8y+8 = \lambda 2y \\ \frac{x^2}{4} + y^2 = 4 \end{cases}$$

$$\frac{2}{x}(2x-2) = \lambda = \frac{8y+8}{2y}$$

$\Downarrow$

$$y(x-1) = x(y+1)$$

$\Downarrow$

$$-y = x$$

$$\frac{x^2}{4} + x^2 = 4$$

$$\Rightarrow \frac{5}{4}x^2 = 4$$

$$\Rightarrow x^2 = \frac{16}{5}$$

$$\Rightarrow x = \pm \frac{4}{\sqrt{5}} \quad y = \mp \frac{4}{\sqrt{5}}$$

$$f\left(\frac{4}{\sqrt{5}}, -\frac{4}{\sqrt{5}}\right) = -2.88854$$

$$f\left(-\frac{4}{\sqrt{5}}, \frac{4}{\sqrt{5}}\right) = 32.88854$$

Ex Find the minimum of $f(x, y, z) = 3x + 2y + z + 5$
under the constraint $9x^2 + 4y^2 - z = 0$

$$\nabla f = (3, 2, 1) \quad \nabla g = (18x, 8y, -1)$$

$$\begin{cases} 3 = \lambda 18x & x = \frac{3}{\lambda \cdot 18} = -\frac{1}{6} \\ 2 = \lambda 8y & y = \frac{2}{8\lambda} = -\frac{1}{4} \\ 1 = -\lambda & \rightarrow \lambda = -1 \\ 9x^2 + 4y^2 - z = 0 & z = 9x^2 + 4y^2 = \frac{9}{36} + \frac{4}{16} = \frac{1}{2} \end{cases}$$

$$f(-\frac{1}{6}, -\frac{1}{4}, \frac{1}{2}) = \frac{9}{2}$$

Note if $y = 0$ then $z = 9x^2$ so $f(x, 0, 9x^2) = 3x + 9x^2 + 5$
goes to $+\infty$ as x goes to $+\infty$. No MAX

Ex 1 Find the minimum of $f(x, y)$ subject to the
constraint $g(x, y) = 0$.

$$a) f(x, y) = x^2 + y^2 \quad g(x, y) = xy - 3 \quad b) f(x, y) = 4x^2 - 4xy + y^2 \\ g(x, y) = x^2 + y^2 - 1$$

$$c) f(x, y, z) = 4x - 2y + 3z \quad g(x, y, z) = 2x^2 + y^2 - 3z$$

Ex 2 Find the maximum volume for the box
with faces parallel to the coordinate axis,
one vertex at $(0, 0, 0)$ and other vertex on plane
 $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ in the first octant.